

Landmarks: Do They Help?

Benchmarking Facial Landmark Streams in Visual Speech Recognition

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Abstract

Visual Speech Recognition (VSR) has been dominated by appearance-only pipelines that feed raw pixel frames into convolutional architectures. In this work we investigate whether facial landmarks — a compact, structured, lower-dimensional representation of lip geometry — carry independent phonetic signal and whether they can complement appearance features in a dual-stream fusion framework. We introduce a dual-stream architecture combining (i) an appearance stream of 3D CNN-extracted spatiotemporal features passed through a BiLSTM, and (ii) a landmark stream processing 50-point lip and chin coordinates (stored as 150-dimensional vectors of x, y, z values) through an MLP frame encoder followed by a Transformer-BiLSTM pipeline. Using the LRW word-level benchmark, we find that landmarks alone reach 62% validation accuracy on a curated Hard-100 subset and **45.06% validation / 44.03% test accuracy** on the full LRW-500 vocabulary — a standalone geometric baseline not previously reported in the literature. Our appearance-only baseline achieves **75.9% validation / 74.9% test accuracy** on LRW-500. Three temporally-aware fusion strategies (temporal concatenation, temporal gated fusion, and cross-attention fusion) are evaluated over pre-extracted frozen representations, converging to a narrow 49–51% test accuracy band, falling below the appearance-only ceiling. These results confirm that landmarks carry non-trivial phonetic signal and that joint end-to-end training is a necessary condition for landmark-guided improvement.

1 Introduction

Visual Speech Recognition (VSR) — the task of recognising spoken words or sentences from lip motion alone — has applications in accessibility aids for deaf and hard-of-hearing individuals, audio-visual restoration, surveillance in acoustically compromised environments, and as an input-processing front-end for

speech-understanding modules in robotics. The field has advanced substantially through the adoption of deep learning, with convolutional and recurrent architectures achieving strong results on established benchmarks such as LRW [1] and LRS2 [2].

Despite this progress, virtually all competitive VSR pipelines operate on a single modality: the raw pixel appearance of the lip region. An alternative, lower-dimensional representation — facial landmarks — has received comparatively little systematic study as a *standalone or complementary* modality. Landmarks are lightweight, geometry-focused, and capture the structural deformation of the lip across time without any dependence on speaker identity, skin tone, or lighting conditions. Whether this representation carries discriminative phonetic signal, and whether it can augment appearance-based models, are empirical questions that remain largely unanswered in the literature.

This paper addresses three specific research questions:

- (1) Do landmarks carry useful phonetic signal as a standalone modality?
- (2) What is the landmark-only benchmark performance on the full LRW-500 benchmark, an evaluation gap in prior work?
- (3) Can a lightweight, resource-efficient model be built by fusing landmark and appearance streams?

Our contributions are as follows. We systematically study the landmark-only pipeline, establishing its architecture through ablation on a curated Hard-100 subset before scaling to the full LRW-500 vocabulary. We report the **first landmark-only standalone classification result on LRW-500**: 44.03% test accuracy. We introduce a dual-stream fusion framework and evaluate three temporally-aware fusion strategies — temporal concatenation, temporal gated fusion, and cross-attention fusion — under a controlled

frozen backbone regime, and characterise the conditions under which landmark features fail to improve appearance-based recognition.

2 Related Work

2.1 Appearance-Based VSR

Early VSR work used hand-crafted features such as PCA coefficients combined with HMM-based temporal modelling [11, 12]. Deep learning pipelines supplanted these, beginning with 2D CNN per-frame feature extraction combined with LSTM temporal modelling [13]. The introduction of 3D convolutional frontends [3] enabled joint spatial and short-range temporal modelling. Afouras et al. [4] established the 3D CNN frontend paired with a 2D ResNet-18 backbone as the dominant visual encoder and demonstrated that Transformer-based temporal models outperform BiLSTM counterparts. This motivated a transition from recurrent to attention-based architectures. The Conformer architecture [14], combining multi-head self-attention with convolutional modules, subsequently became the dominant temporal encoder in state-of-the-art systems. Ma et al. [15] combined a ResNet-18 visual frontend with a Conformer temporal encoder and hybrid CTC-attention decoding, achieving 92% accuracy on LRW.

2.2 Landmark-Based and Geometric Approaches

Facial landmarks have been used as auxiliary alignment signals [9] and in gesture recognition [8]. MediaPipe [5] provides real-time accurate landmark detection, making large-scale feature extraction practical. Recent work revisited geometric representations in the context of cross-speaker generalisation. Xue et al. [16] introduced LipFormer, a Transformer-based visual encoder guided by facial landmarks, improving cross-speaker performance. Wu et al. [17] demonstrated landmark-guided cross-speaker lip reading using dual-stream visual and landmark encoders, showing significant generalisation improvements. However, in these systems the landmark stream is never evaluated independently, leaving its standalone discriminative capacity unquantified on the full LRW-500 benchmark — a gap this work explicitly addresses.

2.3 Multi-Stream Fusion

With multiple input modalities, the key challenge is fusion. Early fusion concatenates features before temporal modelling; late fusion processes each stream

independently before combining predictions. Intermediate approaches use attention mechanisms for selective cross-modal interaction. Jeon et al. [18] proposed a dual-branch Transformer combining Swin Transformers with Conformers for VSR. Teng et al. [19] introduced a two-stage phoneme-level framework fusing visual appearance and facial landmark features, followed by language-model reconstruction. A key gap identified in these works is that each proposes a single fusion mechanism without ablating alternatives under controlled conditions; the extent to which fusion design drives performance gains rather than joint training or backbone capacity remains unestablished. This work addresses that gap directly.

3 Datasets

3.1 LRW (Lip Reading in the Wild)

LRW [1] is a word-level benchmark containing 500 word classes extracted from BBC broadcast television, with approximately 1,000 utterances per word. Each clip spans exactly 29 frames at 25 fps, isolating a 1.16-second temporal window centred on the target utterance. The fixed-length, word-centred structure is a defining characteristic: all classes contain roughly equal sample counts, simplifying training by eliminating the need for variable-length alignment or CTC decoding. The dataset’s “in the wild” nature introduces extensive variability in speaker demographics, lighting, head pose, and background, with over 17,000 unique speaker identities identified. The full training set contains 488,766 clips. We adhere to the official dataset splits, with a validation set of 25,000 clips and a balanced test set of 25,000 clips (50 per class). The 500-word vocabulary spans a diverse range of phonetic structures, from monosyllabic words such as THINK and TRADE to polysyllabic words such as REFERENDUM and WESTMINSTER. LRW is chosen as the primary benchmark because its fixed-length word-level format is well-suited to evaluating and ablating landmark-based geometric features in isolation.

3.2 Hard-100 Subset Construction

To establish and validate the landmark pipeline efficiently before committing to full-scale LRW-500 training, we constructed a 100-word curated subset. Each of the 500 LRW words was scored using its CMU Pronouncing Dictionary [10] representation: words with fewer, more visually ambiguous phonemes received higher difficulty scores. K-Means clustering ($k = 2$) was applied to the resulting score distribution, which

revealed two naturally separable clusters. The top 100 words of the harder cluster were selected as the Hard-100 evaluation set.

This subset served exclusively as a development and ablation environment for the landmark stream (Experiments 001–018 described in Section 6). All appearance stream training and all fusion experiments were conducted on the full LRW-500 vocabulary. The Hard-100 results are presented as ablation context; the primary reported results for all models use the full LRW-500 benchmark.

3.3 LRS2 (Additional Research Scope)

LRS2 [2] is a sentence-level benchmark drawn from BBC programmes, containing over 140,000 variable-length clips with character-level transcriptions. It includes a large Pretrain partition and a smaller Main partition used for fine-tuning. As an additional research effort beyond the core LRW scope, a sentence-level CTC-based appearance stream was developed and trained on LRS2 (see Section 5.5). This component is outside the primary experimental scope of the dual-stream system but is documented here for completeness.

4 Architecture

Our dual-stream system consists of an appearance stream, a landmark stream, and a temporal fusion module. A preprocessing stage common to both streams operates on raw video input.

4.1 Preprocessing

Raw video input first undergoes a shared preprocessing phase. A face detector localises the speaker’s face in each frame, and the lip Region of Interest (ROI) is extracted and spatially normalised.

For the **appearance stream**, each clip is loaded as a sequence of 29 grayscale frames at 96×96 pixel resolution, decoded from .mp4 and pre-stored as NumPy arrays to eliminate per-epoch video decompression overhead during training.

For the **landmark stream**, MediaPipe FaceMesh [5] is applied to each frame to extract 50 facial landmarks (40 lip contour points covering the outer and inner lip boundary, and 10 chin anchor points). Landmarks are normalised per-frame by centring (subtracting the mean coordinate of all landmarks) and scaling by mouth width (the Euclidean distance between the leftmost and rightmost lip corners), making the representation invariant to face size and position across speakers.

Failed detections are zero-filled. All clips are enforced to exactly 29 frames via truncation or zero-padding. Each processed clip is stored as a $(29, 50, 3)$ NumPy array, where the three values per landmark are the x , y , and z coordinates returned by MediaPipe FaceMesh. Flattened per-frame, each landmark frame is therefore a **150-dimensional vector** (50×3). The z coordinate provides a relative depth estimate; it is not on the same metric scale as x and y , a preprocessing limitation acknowledged here.

4.2 Appearance Stream

The appearance stream processes the sequence of 96×96 grayscale frames through a 3D convolutional frontend followed by a BiLSTM temporal model.

A 3D convolutional frontend (Conv3D with kernel $5 \times 7 \times 7$, 64 channels, GroupNorm, ReLU, MaxPool3D) extracts joint spatiotemporal features directly from the raw pixel sequence, capturing both spatial texture and short-range temporal dynamics. A fully connected compression layer then projects the per-frame feature maps into a fixed-dimensional embedding. The resulting per-frame feature sequence is passed through a 2-layer Bidirectional LSTM (512 hidden units total per direction, dropout 0.3), which models longer-range temporal dependencies across the full 29-frame sequence, yielding a 512-dimensional per-timestep representation. For the appearance-only baseline, a linear classification head maps the BiLSTM output to class logits over the 500-word vocabulary.

Note on LRS2 appearance variant: The sentence-level LRS2 variant (Section 5.5) uses a heavier MC3D frontend followed by a pretrained ResNet-18 backbone applied per-frame (with its first convolutional layer replaced to accept the 64-channel frontend output and MaxPool removed), with AdaptiveAvgPool spatial aggregation before the BiLSTM. This heavier variant is not part of the primary LRW-500 pipeline.

4.3 Landmark Stream

The landmark stream takes as input the $(29, 50, 3)$ coordinate sequences produced by preprocessing. Each frame’s 150-dimensional flattened landmark vector is first projected into a 512-dimensional embedding by a 2-layer MLP frame encoder (with ReLU activations), applied independently to each of the 29 frames. This per-frame MLP encodes local lip geometry without any temporal awareness.

A positional encoding is added to the resulting sequence of frame embeddings, after which a multi-head Transformer encoder models global temporal relationships across the full 29-frame sequence via self-attention. The Transformer output is then passed

through a 2-layer BiLSTM, which provides explicit sequential recurrent modelling of articulatory dynamics. The final hidden states are used as the landmark representation. For standalone evaluation, a linear classification head is applied. The final architecture depth and dimensionality were determined by ablation on the Hard-100 subset (Section 6).

4.4 Fusion Module

Following baseline training, the classifier heads are removed from both streams and all backbone parameters are frozen. Temporal feature sequences of shape $(B, 29, 512)$ are extracted from each stream for the full train, validation, and test splits and saved to disk. Fusion layers are trained exclusively on these pre-extracted frozen representations. This design enables a fully controlled comparison of fusion strategies, isolating the contribution of fusion architecture from backbone capacity and joint optimisation effects.

Three temporally-aware fusion strategies are evaluated:

Temporal Concatenation. The per-timestep feature sequences from both streams — each of shape $(B, 29, 512)$ — are concatenated along the feature dimension to form a joint sequence of shape $(B, 29, 1024)$. This joint sequence is passed through a shared 2-layer BiLSTM, and the final hidden state from the last and first timesteps of each direction is concatenated to form a $(B, 1024)$ vector passed to the classification head. This strategy adds 1.8M trainable parameters and provides a direct lower bound on what temporal interaction between streams contributes.

Temporal Gated Fusion. Operating on the full temporal sequences $(B, 29, 512)$ from each stream, independent sigmoid gates $g_t = \sigma(W_g[h_t^a; h_t^l] + b_g)$ are learned to dynamically weight each stream’s contribution at each timestep t , where h_t^a and h_t^l are the appearance and landmark feature vectors at time t respectively. The gated output is then passed through a learned temporal attention pooling mechanism before classification. This variant adds only 2.3M trainable parameters and satisfies the lightweight fusion constraint.

Cross-Attention Fusion. The landmark feature sequence attends over the appearance feature sequence using multi-head cross-attention: landmark features serve as queries (Q) while appearance features provide keys (K) and values (V), enabling the geometric branch to selectively draw on appearance features at relevant timesteps. The cross-attention output is

combined with the landmark stream via a sigmoid residual gate conditioned on both modalities, controlling cross-modal signal retention. A second multi-head self-attention layer performs temporal pooling before classification. This variant adds 6.56M trainable parameters, reflecting the dual attention mechanism, and is compared directly against simpler strategies under identical training conditions.

All three fusion variants are trained with frozen backbones, identical optimiser configurations, identical training budgets of 30 epochs, and identical classification heads, enabling a controlled ablation of fusion design choice.

5 Experimental Setup

5.1 Compute Environment

Experiments were conducted across two compute environments. Early-stage landmark ablation experiments (Hard-100 subset) were run on Google Colab Free Tier using an NVIDIA T4 GPU (16 GB VRAM), which imposed a batch size ceiling of 32 and required crash-safe checkpoint resumption logic to handle session interruptions. Final baseline training and all fusion experiments were conducted on Google Cloud Platform (GCP) using an NVIDIA L4 GPU (24 GB VRAM) on a `g2-standard-8` instance, permitting batch size 64 under mixed-precision FP16 training and eliminating session time limits.

5.2 Training Configuration

The **appearance stream** was trained with batch size 64, AdamW optimiser ($\text{lr} = 1 \times 10^{-4}$, weight decay = 0.01), cosine annealing over 80 epochs, label smoothing 0.1, and gradient clipping at norm 5.0.

The **landmark stream** used separate learning rates for the backbone and classifier head ($\text{lr}_{\text{backbone}} = 3 \times 10^{-5}$, $\text{lr}_{\text{head}} = 1.5 \times 10^{-4}$) under the same cosine annealing schedule over 50 epochs with early stopping (patience = 10), batch size 32 on T4 and 64 on L4.

Fusion layers were trained on the L4 with frozen backbone weights for 30 epochs with AdamW ($\text{lr} = 3 \times 10^{-4}$, weight decay = 0.01) and cosine annealing, adding fewer than 5% additional parameters relative to the frozen encoders across all fusion variants.

5.3 Evaluation Protocol

All models are evaluated on the official LRW test split. The test set contains 25,000 clips across 500 words (50 clips per class), providing a balanced class distribution.

The primary metric is top-1 classification accuracy:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=0}^{N-1} \mathbf{1}[\arg \max f(x_i) = y_i] \quad (1)$$

where $f(x_i)$ denotes the model output logits for clip x_i and y_i is the ground truth class index. Since the test set is perfectly class-balanced, top-1 accuracy is equivalent to macro-averaged accuracy, requiring no per-class weighting. Validation accuracy was monitored throughout training for model selection and early stopping. Final results use the test split exclusively; test performance is consistently within 0.5% variance of the best validation checkpoint. Macro-averaged precision, recall, and F1-score were additionally computed; given the balanced test set, these are numerically close to top-1 accuracy.

5.4 Data Engineering

The raw LRW dataset (70 GB, .mp4) was pre-decoded and stored as NumPy arrays of grayscale frames (≈ 140 GB on disk) to eliminate per-batch video decompression overhead, which was the primary CPU bottleneck during initial training runs. Landmark features were similarly pre-extracted and stored as float16 NumPy arrays of shape $(N, 29, 50, 3)$ per word class on Google Drive. Fusion features were pre-extracted from frozen backbones and saved as sharded .pt files for I/O-efficient loading during fusion training.

5.5 CTC Training on LRS2

As an additional research effort beyond the primary LRW-500 scope, sentence-level CTC-based training was explored on LRS2. A 39-class character-level CTC objective was used. Curriculum learning was applied: training began on clips with transcripts of ≤ 50 characters, expanded to ≤ 100 characters, then to the full dataset. Numerical instability was addressed via a loss ceiling (batches with loss > 1000 were skipped) and a CTC alignment guard (batches where $2L_{\text{target}} + 10 \geq L_{\text{video}}$ were skipped). A three-mode smart resume system (`latest`, `best`, `none`) was implemented to handle Colab session disconnects. Training with the MC3D + ResNet-18 + BiLSTM architecture reached a best validation loss of 2.608 after curriculum pretraining. At fine-tuning stage, decoder output was limited to visually dominant viseme classes, consistent with the known partial viseme-collapse phenomenon requiring validation loss < 2.5 and substantially more compute to overcome. The gap between this result and published LRS2 results is attributable to the physical batch size of 2 (constrained by VRAM), which caused severe BatchNorm instability — the effective batch

size of 16 via gradient accumulation assists only the backward pass weight update, not the forward pass mean and variance computation, which remains extremely noisy at batch size 2. Resolving this requires at least 64 GB VRAM.

6 Ablation Study: Landmark Stream

All landmark ablation experiments in this section were conducted on the Hard-100 subset (Section 3.2) to efficiently identify the optimal architecture before scaling to LRW-500. The Hard-100 results are presented as ablation context; they are not the primary reported results.

6.1 Validation of Landmark Signal (Experiment 001)

Before building a full pipeline, we asked whether landmarks carry any phonetic signal at all. A simple 4-layer Transformer encoder was trained directly on the raw 50-point sequences (150-dimensional per-frame vectors) on the Hard-100 subset. This architecture contains no convolutional component and performs no feature extraction beyond positional encoding. It achieved **61.64% validation accuracy** on Hard-100, confirming that landmarks carry sufficient signal to discriminate 100 word classes well above chance (random baseline: 1%). This result justified continued investment in the landmark stream.

6.2 Classifier Depth: MLP Baselines (Experiments 002–003)

MLPs of varied depths (2, 3, and 4 hidden layers) were evaluated as classifiers on raw landmark features. Critically, mean pooling approaches failed below 7% accuracy: sequences with distinct lip movements collapsed to identical mean vectors, establishing that any approach discarding temporal ordering is fundamentally unsuitable for lip reading. Deeper MLPs with temporal context improved training accuracy but showed limited generalisation, motivating the addition of explicit temporal modelling.

6.3 Adding Temporal Context: BiLSTM (Experiment 004)

Appending a 2-layer BiLSTM between the feature encoder and the classifier produced a measurable improvement in validation accuracy over MLP-only baselines, confirming the critical importance of sequential modelling for capturing articulatory dynamics.

6.4 Transformer Integration (Experiments 005–006)

Prepending a Transformer encoder (4 layers, 4 attention heads) before the BiLSTM produced the best results in this group. Experiment 006 established the first complete MLP–Transformer–BiLSTM landmark pipeline, which became the basis for all subsequent experiments.

6.5 Augmentation (Experiments 007–008)

Coordinate jitter and random temporal subsampling were evaluated as augmentation strategies. Standard augmentations (flipping, cropping, noise) significantly degraded accuracy, since landmark coordinates carry dense semantic meaning in their precise spatial positions and perturbations corrupt the articulatory signal rather than regularising the model. Moderate jitter improved generalisation marginally; aggressive subsampling degraded performance, suggesting the model relies on dense temporal coverage of lip motion.

6.6 Architecture Granularity (Experiments 009–014)

A grid search over Transformer depth (2, 4, 6 layers), positional encoding variants (sinusoidal vs. learned), and output projection dimension (128, 256, 512) was conducted. Key findings:

- 4-layer Transformer depth was optimal; deeper models overfit on Hard-100.
- Sinusoidal positional encoding marginally outperformed learned encoding.
- 512-dimensional per-frame projections balanced capacity and generalisation.
- Positional encoding *prior to* the BiLSTM degraded performance substantially, confirming that explicit recurrent temporal modelling is necessary and cannot be replaced by positional signals alone.

Finalised Architecture (Experiment 010). The finalised landmark pipeline — 2-layer MLP frame projector (150→512), 4-layer Transformer encoder, 2-layer BiLSTM, linear classifier — achieved **61.58% validation accuracy** on the Hard-100 subset. The near-identical result between the 4-layer Transformer-only baseline (61.64%, Experiment 001) and the full MLP–Transformer–BiLSTM pipeline (61.58%, Experiment 010) suggests that the Hard-100 task has an architecture-agnostic ceiling near 62%, likely imposed by the intrinsic expressiveness limit of sparse geometric keypoints rather than model capacity.

6.7 Scaling to Full LRW-500 Vocabulary (Experiments 015–018)

The finalised pipeline was extended to the full 500-word LRW vocabulary, with and without warm-starting from Hard-100 checkpoint weights.

Table 1: Landmark-only accuracy on LRW (ablation scaling results). *These are development-phase results; final reported results use the full training configuration detailed in Table 2.*

Vocabulary	Warm-start	Val Acc (%)
Hard-100	–	61.6
LRW-500	No	40.8
LRW-500	Yes	42.0

Warm-starting from the Hard-100 checkpoint provided a consistent improvement of approximately 1.2 percentage points over training from scratch on LRW-500, confirming positive transfer from the curated subset. The final fully-trained landmark-only model (reported in Section 7) achieves 45.06% validation / 44.03% test accuracy on LRW-500.

7 Results

Table 2 presents the complete experimental results for all five models evaluated on the official LRW-500 test split.

Table 2: Complete results on LRW-500. Val/Test Acc from best validation checkpoint and final test split. †Fusion-layer params only; backbones frozen.

Model	Params	Ep.	Val	Test
Landmarks-only (MLP+Transf.+BiLSTM)	4.05M	80	45.06%	44.03%
Appearance-only (3D CNN+BiLSTM)	4.1M	80	75.90%	74.90%
Temp. Concat. Fusion	1.8M [†]	30	50.77%	50.76%
Temp. Gated Fusion	2.3M [†]	30	49.41%	49.33%
Cross-Attention Fusion	6.56M [†]	30	50.12%	49.05%

7.1 Landmark-Only Baseline

The landmark-only pipeline achieves **45.06% validation accuracy** and **44.03% test accuracy** on LRW-500. To the best of our knowledge, this constitutes the first reported standalone landmark-only classification result on the full LRW-500 benchmark. Existing systems that incorporate facial landmarks [16, 17] use them exclusively as auxiliary guidance within

jointly-trained multi-stream architectures; the landmark stream is never evaluated independently.

The training curves (Figure ??) show a continuously widening generalisation gap: validation loss plateaus from epoch 10 while training loss continues to decrease, confirming that the ceiling reflects a fundamental limit of the geometric feature space rather than insufficient training. Additional epochs yield no validation improvement regardless of decreased training loss.

While substantially below the appearance-only baseline, 44.03% meaningfully exceeds early handcrafted approaches using PCA and HMM, which achieved well below 20% on comparable vocabulary sizes [11].

7.2 Appearance-Only Baseline

The appearance-only baseline achieves **75.90% validation accuracy** and **74.90% test accuracy** on LRW-500, establishing the appearance stream as the dominant modality by a margin of approximately 31 percentage points over landmarks alone. Validation accuracy rises sharply in the first 15 epochs before gradually plateauing at 74.89% by epoch 66. The persistent train-validation gap (89.2% train vs. 74.9% validation, a 14.3 pp gap) indicates the model has more capacity than the training data can fully regularise; further gains would likely require a stronger backbone such as ResNet-18 or additional regularisation. This result is competitive with published BiLSTM-based architectures of comparable scale reported by Afouras et al. [4] (70–76% range), achieved without large-scale pre-training.

7.3 Fusion Results

All three fusion strategies converge to a narrow test accuracy band of 49.05–50.76% despite significant differences in architectural complexity and parameter count. Temporal concatenation achieves the best result at **50.76% test accuracy** with only 1.8M fusion parameters. Cross-attention delivers 49.05% despite deploying 6.56M parameters — nearly four times the parameter count of concatenation — for no corresponding accuracy benefit. All fusion variants fall substantially below the appearance-only baseline and all exceed the landmark-only baseline.

Critically, all fusion variants exhibit large train-validation gaps: gated fusion reaches 94.7% training accuracy while validation accuracy plateaus near 49%, and cross-attention reaches 91% training accuracy with similar validation behaviour. Temporal concatenation is the exception, showing tightly coupled train/val curves throughout all 30 epochs. The large gaps in the other two variants are diagnostic:

the fusion layers overfit to the fixed feature distribution rather than learning generalisable interaction structure.

7.4 Per-Class Analysis

Per-class accuracy exhibits a bimodal rather than uniform distribution across the 500-word vocabulary: a large cluster of words achieves 50–90% accuracy, but approximately 90 words sit near 0%, indicating systematic failure on a subset of the vocabulary rather than uniformly degraded performance. Mean accuracy across all classes is 49.1%.

Words with visually distinctive mouth shapes and salient movement trajectories are recognised reliably across all model variants: WESTMINSTER (96%), REFERENDUM (96%), and INFORMATION (94%). By contrast, the failure cluster is dominated by visually underspecified words — ABOUT, ACCORDING, AFTER — whose lip patterns are subtle or nearly identical to neighbouring vocabulary members.

The most frequent confusion pairs are BIL-LION/MILLION, CLAIMS/GAMES, and CONFERENCE/OFTEN, which share viseme sequences rather than semantic category. This confirms that residual errors are phonetically principled and attributable to homophemy and near-homophemy rather than model failure — a pattern consistent with the broader lip reading literature.

8 Comparative Analysis with Prior Work

8.1 Landmark-Only Benchmark on LRW-500

To the best of our knowledge, no prior work reports a standalone landmark-only classification result on the full LRW-500 benchmark. The 44.03% test accuracy reported here constitutes the first published coordinate-based geometric baseline on this benchmark, providing a reference point for future work in geometric VSR. The result significantly exceeds early handcrafted approaches while confirming that geometric features alone are substantially weaker than pixel-level appearance for large-vocabulary word classification.

8.2 Fusion Strategy Comparison

The three fusion strategies evaluated here converge to a 49–51% accuracy band, demonstrating that fusion strategy selection has minimal impact under the frozen backbone regime. This is itself a key finding: Wu et al. [17] and Xue et al. [16] each propose a single fusion mechanism without ablating alternatives,

making it difficult to attribute their reported gains to fusion design specifically versus joint training. Our controlled comparison directly answers this question: architectural sophistication in the fusion layer does not compensate for the absence of joint optimisation.

8.3 Parameter Efficiency

A deliberate design constraint of this work was to maintain lightweight fusion layers. The temporal concatenation fusion achieves the best result at 50.76% with only 1.8M fusion parameters. The cross-attention variant delivers minimal improvement despite nearly four times the parameter count. The appearance-only baseline at 4.1M total parameters and 74.9% accuracy represents a strong efficiency–accuracy trade-off for a from-scratch lightweight system, comparable to the 70–76% range reported by Afouras et al. [4] for BiLSTM-based architectures of similar scale. State-of-the-art systems with Auto-AVSR-class backbones [15] operate with substantially larger models and large-scale pre-training, making them unsuitable for constrained deployment settings.

8.4 Do Landmarks Help Appearance?

The results provide a clear answer under the conditions tested: **under a frozen backbone constraint, landmark features do not improve upon appearance-only recognition.** The best fusion result of 50.76% falls 24.1 pp below the appearance-only baseline of 74.9%. Fusing landmark features with a frozen pre-trained appearance model therefore *actively degrades* performance relative to appearance alone. This contrasts with gains reported by Wu et al. [17] and Xue et al. [16], both of whom demonstrate landmark-guided improvements under end-to-end joint training. The discrepancy indicates that landmarks provide complementary value only when the appearance encoder is optimised to attend to geometrically salient regions — which requires joint training. The practical implication is direct: incorporating landmarks as a post-hoc signal over a frozen pre-trained appearance model is unlikely to yield gains.

9 Discussion

9.1 Architecture-Agnostic Ceiling on Landmarks

The near-identical validation accuracy of the 4-layer Transformer-only model (61.64%, Experiment 001) and the full MLP–Transformer–BiLSTM pipeline (61.58%, Experiment 010) on Hard-100 suggests that

the Hard-100 ceiling is not architectural but representational. Across all architectures tested — ViT variants, deeper Transformers, wider MLPs — the landmark stream plateaued near 62% on LRW-100 and near 44% on LRW-500. This confirms the ceiling is an intrinsic property of geometric features rather than a model design issue. The 50-point landmark set, despite covering the lip and chin region, may lack the expressiveness to distinguish consonant classes that differ primarily in fine-grained spatiotemporal patterns below the resolution of sparse keypoints. Denser landmark sets or auxiliary features encoding lip curvature, inter-landmark velocity, or delta-delta acceleration may push this ceiling higher.

9.2 The Bottleneck is Representational, Not Architectural

The narrow convergence of all three fusion strategies to a 49–51% accuracy band, across temporal concatenation (1.8M params), gated fusion (2.3M params), and cross-attention (6.56M params), confirms that fusion architecture choice is not the performance bottleneck. The frozen appearance and landmark encoders were each trained independently with no signal from the other modality; neither learned to produce representations that are complementary. The fusion layer must reconcile representations that were never incentivised to interact, which limits the expressiveness of any fusion design regardless of its capacity.

9.3 Lightweight Design

A secondary goal was to build a resource-efficient model. The landmark stream is substantially lighter than the appearance stream: it operates on 150-dimensional per-frame inputs versus full-resolution image tensors, requires no convolutional frontend, and runs on CPU-extracted MediaPipe keypoints at extraction time. The entire landmark-only pipeline (4.05M parameters) operates on pre-extracted geometric features, making it attractive for edge deployment or as a low-power fallback modality. The appearance-only baseline (4.1M parameters, 74.9% test accuracy) presents the strongest efficiency–accuracy trade-off from this work.

9.4 Resource Constraints

All primary experiments were conducted under significant compute constraints. The Colab T4 (16 GB VRAM) imposed a physical batch size ceiling for early landmark experiments, and the LRS2 additional-scope experiments were severely limited by the batch size of

2 imposed by the larger MC3D+ResNet-18 architecture. The transition to GCP L4 (24 GB VRAM) for final baseline and fusion training enabled batch size 64 and was necessary to complete the full LRW-500 training runs. Published state-of-the-art LRW results were achieved on multi-GPU A100/V100 clusters; the gap between our results and SOTA is attributable to compute budget rather than architectural choice.

10 Conclusion

We presented a systematic study of facial landmarks as a modality for Visual Speech Recognition. Our findings demonstrate that:

- (1) The landmark stream (MLP frame projector on 150-dimensional per-frame vectors + Transformer encoder + BiLSTM) achieves 62% validation accuracy on the Hard-100 curated ablation subset and **45.06% validation / 44.03% test accuracy on the full LRW-500 benchmark** — the first reported coordinate-based geometric baseline on this dataset.
- (2) The appearance-only baseline (3D CNN frontend + BiLSTM) achieves **75.90% validation / 74.90% test accuracy on LRW-500**, establishing a strong from-scratch lightweight ceiling.
- (3) Three temporally-aware fusion strategies (temporal concatenation, temporal gated fusion, cross-attention) converge to a narrow 49–51% test accuracy band under frozen backbone conditions — all falling substantially below the appearance-only ceiling — confirming that fusion architecture choice is not the performance bottleneck.
- (4) Landmark features do not improve appearance-based recognition under a frozen backbone regime, and actively degrade performance. Joint end-to-end training is a necessary condition for landmark-guided improvement, consistent with findings in [16, 17].

The landmark stream carries genuine phonetic signal (44.03% standalone, far above chance on 500 classes). However, realising the complementary potential of that signal in a fused system requires joint optimisation. Future work directions include: (1) full end-to-end joint training of both streams, initialising from the independently trained checkpoints produced in this work; (2) adopting a stronger ResNet-18 appearance backbone to close the gap to state-of-the-art; (3) scaling to sentence-level recognition on LRS2 with CTC or hybrid CTC-attention decoding;

(4) incorporating viseme-aware supervision to address homophemy-driven confusion; and (5) augmenting the landmark input with first- and second-order delta (velocity and acceleration) features.

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